Optimising Blockchain Voting with Zero-Knowledge Rollup Layer-2

Dissertation Proposal Presentation

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December 2025

Aim of Project

Context

Traditional electronic voting lacks transparency and verifiability; blockchain offers decentralised security but faces scalability, privacy, and cost challenges.

Issues to Address

- **Privacy–Transparency Dilemma:** Public blockchains expose transaction data, conflicting with ballot secrecy
- **Scalability Constraints:** Layer-1 throughput (15–30 TPS) insufficient for national elections
- **High Costs:** L1 transactions exceed $1.10; storing 500 voters costs ∼10M gas

Proposed Solution

Integrate **Zero-Knowledge Proofs (ZKPs)** within a **zk-Rollup Layer-2** architecture.

Background: The Problem Space

Blockchain Voting Challenges

- Centralised e-voting: Single points of failure, DDoS vulnerability
- Layer-1 limitations:
  - 10M votes at 15 TPS = **7.7 days**
  - Max ∼40 votes per contract before gas limit
- Public transparency conflicts with secret ballot

The Solution Components

- **Zero-Knowledge Proofs:** Prove eligibility & ballot validity without revealing identity
- **zk-Rollups:** Off-chain execution with cryptographic proofs to L1
- **Result:** Inherit L1 security with L2 scalability and privacy

Privacy Risk: Public Voter Identity

ETH Price: $2,929.04 (-0.25%) Gas: 0.028 Gwei

Search by Address / Txn Hash / Block / Token / Domain Name

Block:	21751226 2369743 Block Confirmations
Timestamp:	331 days ago (Feb-01-2025 10:59:35 AM UTC) | Confirmed within 7 secs
Sponsored:	Advertise on Scrollscan
From:	eventhorizoncommunity.eth (0xb35659cbac913d5e4119f2af47fd490a45e2c826) eventhorizoncommunit...
To:	0xc0Da02939E1441F497fd74F78cE7Decb17B66529 (Compound: Governance)
Value:	0 ETH ($0.00)
Transaction Fee:	0.0002647762335294 ETH $0.78
Gas Price:	3.201647322 Gwei (0.000000003201647322 ETH)
Ether Price:	$3,116.61 / ETH
Gas Limit & Usage by Txn:	113,611 | 82,700 (72.79%)
Gas Fees:	Base: 2.728656753 Gwei | Max: 4.120906292 Gwei | Max Priority: 0.472990569 Gwei
Burnt & Txn Savings Fees:	Burnt: 0.0002256599134731 ETH ($0.66) Txn Savings: 0.000076022716819 ETH ($0.22)

Privacy Risk: Public Vote Details

Ether Price:	$3,116.61 / ETH
Gas Limit & Usage by Txn:	113,611 | 82,700 (72.79%)
Gas Fees:	Base: 2.728656753 Gwei | Max: 4.120906292 Gwei | Max Priority: 0.472990569 Gwei
Burnt & Txn Savings Fees:	Burnt: 0.0002256599134731 ETH ($0.66) Txn Savings: 0.000076022716819 ETH ($0.22)
Other Attributes:	Txn Type: 2 (EIP-1559) Nonce: 216 Position In Block: 40

Input Data:

#	Name	Type	Data
0	proposalId	uint256	393
1	support	uint8	1
2	reason	string	The Event Horizon Community Voted to Support this Proposal ehCOMP-164: EventHorizon.vote/vote/compound/ehCOMP-164

Switch Back | View In Decoder

Layer-2 Fees: Cost Reduction Through Rollups

L2 Fees

Ethereum Layer-1 is expensive.
How much does it cost to use Layer-2?

All L2s | Full Rollups

Name	Send ETH	Swap tokens
Metis Network ⚠	$0.04	$0.18
Loopring	$0.04	$0.59
zkSync Era	$0.07	-
zkSync Lite	$0.09	$0.22
Optimism	$0.09	$0.18
Arbitrum One	$0.09	$0.27
Boba Network	$0.15	$0.17
DeGate	$0.16	$0.18
StarkNet	$0.19	$0.57
Polygon zkEVM	$0.19	$2.75
Ethereum	$1.10	$5.48

How can rollups reduce their fees?
Read our first blog-post "Crunching the Calldata".

Research Question 1

Research Question

How can zero-knowledge proofs combined with a zk-rollup Layer-2 architecture improve the **scalability**, **voter privacy**, and **verifiability** of blockchain-based electronic voting systems?

Motivation

Current Layer-1 blockchain voting systems cannot support national-scale elections due to limited throughput, high costs, and insufficient privacy guarantees.

Research Question 1: Approach

Methodology

Design and implement **ZK-Verified Election Rollup (ZKER)** on Polygon zkEVM:

- L2 sequencer batches vote transactions
- Aggregator generates zero-knowledge validity proofs
- Proof submitted to an L1 verifier contract

Expected Outcomes

- Measure **batch size** and **votes per proof**
- Quantify **data compression ratio** (L2 vs L1)
- Evaluate **proof verification success rates**

Research Question 2

Research Question

How effectively can zero-knowledge proofs guarantee **voter anonymity** and **ballot secrecy** without compromising end-to-end verifiability?

Motivation

Elections require secret ballots, yet voters must be authenticated as eligible and prevented from double-voting without revealing their identity.

Research Question 2: Privacy-Preserving Voting Protocol

Methodology: Three-Phase Protocol

1. **Authentication:** Voter proves eligibility using a Merkle tree commitment and a one-time nullifier to prevent double voting.
2. **Ballot Encryption:** Vote is encrypted and accompanied by a zero-knowledge proof to ensure ballot validity without revealing its content.
3. **Tallying:** Encrypted ballots are aggregated and decrypted collectively using threshold decryption to reveal only the final result.

Expected Outcomes

- Nullifiers prevent double voting while preserving anonymity.
- Any observer can verify the final tally using only public data (ZK proofs, aggregated ciphertext, and DKG output).

Research Question 3

Research Question

To what extent does a zk-rollup-based voting system **reduce transaction costs** and **increase throughput** compared to a traditional Layer-1 blockchain voting model?

Motivation

Ethereum Layer-1 fees (typically $1.10 or higher) and limited throughput make large-scale blockchain voting economically and technically impractical.

Research Question 3: Performance Evaluation

Methodology: Comparative Benchmarking

- **Environment A:** ZK-Verified Election Rollup (ZKER) on Polygon zkEVM
- **Environment B:** Baseline voting on Ethereum Layer-1 (simulated)
- **Test Setup:** Inject identical high-volume batches (e.g., 10,000 votes per batch)

Expected Outcomes

- Compare achieved **throughput (TPS)**
- Compare **average cost per vote**
- Target throughput: >70 TPS
- Expected L2 fee: <$0.10 per vote

Research Question 4 (Nice to Have)

Research Question

Which zero-knowledge proof scheme (**zk-SNARKs**, **zk-STARKs**, **Bulletproofs**) provides the optimal balance of performance and security for a Layer-2 rollup voting system?

Motivation

Different zero-knowledge proof systems present trade-offs in efficiency, trust assumptions, and long-term security that directly affect rollup-based voting designs.

Research Question 4: Intended Comparison

Methodology

Conduct a literature-based comparative analysis of zk-SNARKs, zk-STARKs, and Bulletproofs within the context of Layer-2 rollup voting systems.

Expected Outcomes

Identify and compare the proof systems based on:

- Proof size and on-chain data requirements
- Verification cost and computational efficiency
- Trusted setup assumptions
- Security properties, including post-quantum resistance

Research Question 5 (Nice to Have)

Research Question

How does a zero-knowledge voting system behave across different **EVM-compatible zk-rollup environments** (e.g., Polygon zkEVM vs zkSync Era)?

Motivation

Architectural differences between zk-rollup platforms may significantly affect voting system performance, cost efficiency, and security guarantees.

Research Question 5: Cross-Platform Evaluation

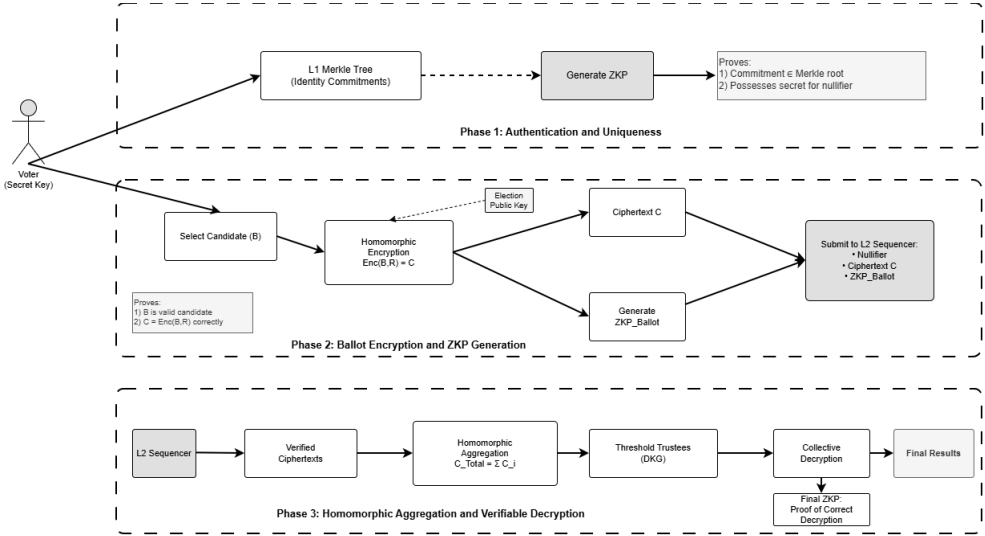
Methodology

- Primary evaluation: ZK-Verified Election Rollup (ZKER) proof-of-concept on **Polygon zkEVM**
- Comparative evaluation: **zkSync Era** assessed using existing literature and public benchmarks
- Analyse architectural design choices that influence e-voting key performance indicators (KPIs)

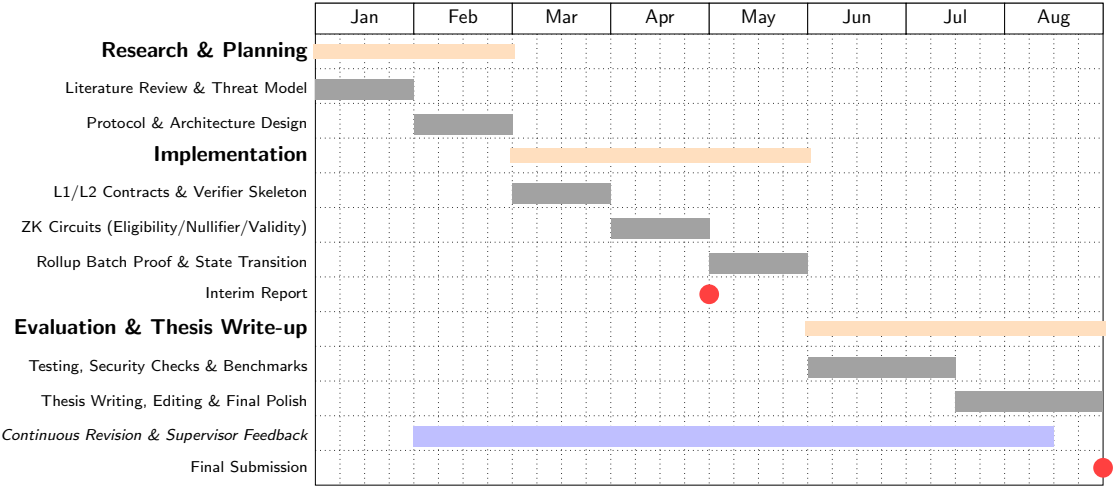
Expected Outcomes

- Compare **transaction costs**: average Layer-2 fees and Layer-1 proof verification overhead
- Compare **scalability**: reported throughput and batch processing capabilities
- Identify architectural features best suited to secure and scalable voting systems

Proposed Architecture



Project Plan



Technology Stack

Smart Contracts & Blockchain

- **Language:** Solidity
- **Framework:** Hardhat
- **L1 Network:** Ethereum Sepolia testnet
- **L2 / Rollup:**
  - zkSync Era testnet
  - Polygon zkEVM testnet

Zero-Knowledge Proofs

- **ZK Tools:** Circom + snarkjs + Groth16

Backend

- **Language:** Node.js / TypeScript

Frontend & Testing

- **UI:** React.js
- **Wallet Integration:** MetaMask
- **Contract Tests:** Hardhat / Jest

Main Project Uncertainties

- **Realistic workload modelling:** Simulating realistic election workloads (voter counts, turnout patterns) may be challenging.
- **Testnet gas and faucets:** Limited faucet funds and unstable testnet faucets may restrict.
- **Imperfect privacy:** Miners may still infer who voted for what by correlating on-chain activity with off-chain metadata such as addresses, timing, network information.

Thank you for listening

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